Injury 57 (2026) 112564

Contents lists available at ScienceDirect

Injury

journal homepage: www.elsevier.com/locate/injury

Concern on "Ultrasound-guided percutaneous cryoneurolysis of intercostal nerves in traumatic rib fractures"

Dear Editor,

We read the article by Forrester et al., recently published in Injury, with great interest [1]. We applaud the authors for their remarkable work. However, we have some concerns regarding the manuscript and would like to highlight them through this correspondence.

In this prospective study, Forrester et al. have tried to evaluate the role of cryoneurolysis of the intercostal nerves among patients with traumatic rib fractures [1]. Our major concern in the index study is regarding the effectiveness of the technique in pain reduction and the interpretation of results. From Figure 2, it is evident that 3 patients had no change in pain at discharge (baseline pain of 10/10 NRS), and 2 patients developed increased pain at discharge (from baseline 8/10 to 10/10 NRS) [1]. This can be due to the failure of the technique, though the authors have not defined failure in the methodology [1,2] A failure rate of 5/15 (35 %) is high and raises questions about the effectiveness of the technique for pain reduction. This is also evident by a median pain score of 7 (NRS of 7 equals moderate to severe pain) at the time of discharge [1]. The authors should have mentioned the post-procedure pain at least two or three time point (e.g. at 1 hour, 6 h or 12 h), which could have better depicted the pain relief following the procedure. Though authors are interested in multimodal analgesia, they have not mentioned rescue analgesia or how they give additional analgesia [1,2]. The readers would be interested to know whether they had a fixed analgesia regimen, or if it was based on measured pain score in NRS.

Our next concern is the author's preference of cryoneurolysis over other conventional analgesia regimens [1]. As per the published literature, it is not novel to use cryo-neurolysis to manage the post-surgical pain following thoracic pathology, but thoracic epidural remains the gold standard treatment to manage such pain [3,4]. Cryoneurolysis is primarily used to manage chronic pain refractory to conventional analgesics, however, researchers have evaluated it as an adjunct along with the epidural and sole treatment modality for managing acute painful conditions, with varying results [5]. Although cryoneurolysis has shown its promising role but disadvantages like unpredictable duration of analgesia, requirement of repeated application after 3 months or when pain recurs, and side effects like muscle weakness have limited its use in the current practices [6]. So, a discussion on the available analgesia options like thoracic epidural, paravertebral block, and erector spinae plane block (ESPB) would have been more informative for the readers [3,4,7]. Among the options, ultrasound-guided ESPB is promising for pain management associated with multiple ribs fracture due to its ease of administration and safety profile [7–9]. Even a single short ESPB provides sustained pain relief and adding a catheter can provide a reliable long-term pain relief [7,8]. As per our experience, a single short ESPB of 25 ml of ropivacaine 0.375 % in patients with multiple ribs fracture provide a pain score of $\leq$ 2/10 at 30 days follow-up time. Additionally, the author's comment in the discussion that cryoneurolysis contributes to the reduction in length of hospital stay, time to functional recovery, and reduction in the overall healthcare cost related to rib fracture seems to be an overstatement not supported by scientific proof [1].

In the methods, the authors have excluded patients with Glasgow coma scale score (GCS) <13. We are curious to know how the authors calculated the pain intensity in NRS among patients with GCS 13 and 14 and in patients having intracranial hemorrhage (Table 1)? [1]. From Table 1, it seems that 55 % of patients had associated injuries along with chest injury, with a median rib fracture count of 5, which accounts for an abbreviated injury scale (AIS) score of 3 or 4. Considering that 8 patients had associated injury like intracranial injury, solid organ and pelvic injury, it seems the authors have undercalculated the injury severity score (ISS) than expected [1,10]

Lastly, the authors have administered 2 ml of local anesthetic (LA) for hydro-dissection, however, the details of LA, including type and concentration not mentioned [1]. Multisystem trauma patients would be more prone to local anesthetic systemic toxicity (LAST) and consider the multiple levels of injections required for intercostal block and also since systemic levels are highest after intercostal block, utmost care is needed for the dose and type of LA used [11,12]. Preparation for LAST, including intralipid availability, airway management, and resuscitation backup is essential [11].

To conclude, we appreciate the authors for their diligent efforts and hope that the above-mentioned points will increase the authenticity of their valuable work.

Author Contributions Statement (CRedIT Statement):

CRM, AKB, and AG: Conceptualization (lead); Methodology (lead); Software and investigation (lead); Writing – original draft (lead); Resources; Project administration; Writing – review and editing (equal). RVR and AVP: Writing – original draft (lead); Software and investigation (support); visualization; Writing – review and editing. CRM did the overall supervision of the whole study, and all authors made a substantial contribution. All authors have read and agreed to the content of the final manuscript [AKB: Amiya Kumar Barik, AG: Anju Gupta, CRM: Chitta Ranjan Mohanty, RVR: Rakesh Vadakkethil Radhakrishnan, AVP: Aditya Vikram Prusty]

Consent to Participate: Written consent was not needed as it is a reply letter to one of the published articles in the Journal.

Ethical Approval: Not applicable

https://doi.org/10.1016/j.injury.2025.112564
Accepted 22 June 2025
Available online 26 June 2025
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Ethics statement

The manuscript is a reply comment to one of the published articles in the journal and does not involve any human or animal participants. Hence the institutional review board approval was not required.

CRediT authorship contribution statement

Amiya Kumar Barik: Conceptualization, Writing – original draft, Writing – review & editing. **Anju Gupta:** Conceptualization, Writing – original draft, Writing – review & editing. **Chitta Ranjan Mohanty:** Conceptualization, Writing – original draft, Writing – review & editing. **Rakesh Vadakkethil Radhakrishnan:** Resources, Writing – original draft, Writing – review & editing. **Aditya Vikram Prusty:** Software, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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